

Frequencies

[iomt_data] D:\IoMT-Encryption-Simulation\4_spss_outputs\output\iomt_encryption_analysis_ready.sav

Statistics

Encryption condition code

N	Valid	40
	Missing	0

Encryption condition code

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Unencrypted	10	25.0	25.0	25.0
	ECC	15	37.5	37.5	62.5
	RSA-SHE	15	37.5	37.5	100.0
	Total	40	100.0	100.0	

Means

Case Processing Summary

	Included		Cases Excluded		Total	
	N	Percent	N	Percent	N	Percent
Output file size in kilobytes * Encryption condition code	40	100.0%	0	0.0%	40	100.0%
Average row length in bytes * Encryption condition code	40	100.0%	0	0.0%	40	100.0%
Measured encryption time in seconds * Encryption condition code	40	100.0%	0	0.0%	40	100.0%
Clear-text exposure percentage * Encryption condition code	40	100.0%	0	0.0%	40	100.0%

Report

Encryption condition code		Output file size in kilobytes	Average row length in bytes	Measured encryption time in seconds	Clear-text exposure percentage
Unencrypted	N	10	10	10	10
	Mean	61.64150	61.02590	.00000000	100.0000
	Std. Deviation	.005922	.006064	.00000000	.00000
	Minimum	61.634	61.018	.000000	100.00
	Maximum	61.650	61.035	.000000	100.00
ECC	N	15	15	15	15
	Mean	1825.35319	428.00000	.06399869	.0000
	Std. Deviation	1186.218687	.000000	.039449822	.00000
	Minimum	839.937	428.000	.030559	.00
	Maximum	4199.312	428.000	.142190	.00
RSA-SHE	N	15	15	15	15
	Mean	35791.88965	3104.00000	2.92742547	.0000
	Std. Deviation	12089.403372	.000000	.974129258	.00000
	Minimum	18199.312	3104.000	1.494043	.00
	Maximum	60664.155	3104.000	4.921349	.00
Total	N	40	40	40	40
	Mean	14121.87644	1339.75648	1.12178406	25.0000
	Std. Deviation	18504.789715	1391.451420	1.532381538	43.85290
	Minimum	61.634	61.018	.000000	.00
	Maximum	60664.155	3104.000	4.921349	100.00

Explore

Encryption condition code

Case Processing Summary

		Cases			
		Valid		Missing	
	Encryption condition code	N	Percent	N	Percent
Output file size in kilobytes	Unencrypted	10	100.0%	0	0.0%
	ECC	15	100.0%	0	0.0%
	RSA-SHE	15	100.0%	0	0.0%
Average row length in bytes	Unencrypted	10	100.0%	0	0.0%
	ECC	15	100.0%	0	0.0%
	RSA-SHE	15	100.0%	0	0.0%
Measured encryption time in seconds	Unencrypted	10	100.0%	0	0.0%
	ECC	15	100.0%	0	0.0%
	RSA-SHE	15	100.0%	0	0.0%

Case Processing Summary

	Encryption condition code	Cases	
		N	Percent
Output file size in kilobytes	Unencrypted	10	100.0%
	ECC	15	100.0%
	RSA-SHE	15	100.0%
Average row length in bytes	Unencrypted	10	100.0%
	ECC	15	100.0%
	RSA-SHE	15	100.0%
Measured encryption time in seconds	Unencrypted	10	100.0%
	ECC	15	100.0%
	RSA-SHE	15	100.0%

Descriptives

Encryption condition code		Statistic	
Output file size in kilobytes	Unencrypted	Mean	61.64150
		95% Confidence Interval for Mean	Lower Bound
			Upper Bound
		5% Trimmed Mean	61.64144
		Median	61.64111
		Variance	.000
		Std. Deviation	.005922
		Minimum	61.634
		Maximum	61.650
		Range	.017
		Interquartile Range	.010
		Skewness	.379
		Kurtosis	-.977
	ECC	Mean	1825.35319
		95% Confidence Interval for Mean	Lower Bound
			Upper Bound
		5% Trimmed Mean	1748.21199
		Median	1259.85840
		Variance	1407114.774
		Std. Deviation	1186.218687
		Minimum	839.937
		Maximum	4199.312
		Range	3359.375
		Interquartile Range	1931.641
		Skewness	1.037

Descriptives

Encryption condition code		Std. Error	
Output file size in kilobytes	Unencrypted	Mean	.001873
		95% Confidence Interval for Mean	Lower Bound
			Upper Bound
		5% Trimmed Mean	
		Median	
		Variance	
		Std. Deviation	
		Minimum	
		Maximum	
		Range	
		Interquartile Range	
		Skewness	.687
		Kurtosis	1.334
	ECC	Mean	306.280348
		95% Confidence Interval for Mean	Lower Bound
			Upper Bound
		5% Trimmed Mean	
		Median	
		Variance	
		Std. Deviation	
		Minimum	
		Maximum	
		Range	
		Interquartile Range	
		Skewness	.580

Descriptives

Encryption condition code		Statistic	
Average row length in bytes	RSA-SHE	Kurtosis	-.542
		Mean	35791.88965
		95% Confidence Interval for Mean	Lower Bound 29097.00121
			Upper Bound 42486.77808
		5% Trimmed Mean	35387.46256
		Median	36398.53027
		Variance	146153673.88
		Std. Deviation	12089.403372
		Minimum	18199.312
		Maximum	60664.155
		Range	42464.844
		Interquartile Range	18199.219
		Skewness	.584
		Kurtosis	-.096
	Unencrypted	Mean	61.02590
		95% Confidence Interval for Mean	Lower Bound 61.02156
			Upper Bound 61.03024
		5% Trimmed Mean	61.02583
		Median	61.02550
		Variance	.000
		Std. Deviation	.006064
		Minimum	61.018
		Maximum	61.035
		Range	.017
		Interquartile Range	.011
		Skewness	.379
		Kurtosis	-.977
	ECC	Mean	428.00000
		95% Confidence Interval for Mean	Lower Bound 428.00000
			Upper Bound 428.00000
		5% Trimmed Mean	428.00000
		Median	428.00000
		Variance	.000
		Std. Deviation	.000000
		Minimum	428.000
		Maximum	428.000
		Range	.000
		Interquartile Range	.000
		Skewness	.

Descriptives

Encryption condition code		Std. Error	
Average row length in bytes	RSA-SHE	Kurtosis	1.121
		Mean	3121.470528
		95% Confidence Interval for Mean	Lower Bound
			Upper Bound
		5% Trimmed Mean	
		Median	
		Variance	
		Std. Deviation	
		Minimum	
		Maximum	
		Range	
		Interquartile Range	
		Skewness	.580
		Kurtosis	1.121
		Mean	.001917
		95% Confidence Interval for Mean	Lower Bound
			Upper Bound
Average row length in bytes	Unencrypted	5% Trimmed Mean	
		Median	
		Variance	
		Std. Deviation	
		Minimum	
		Maximum	
		Range	
		Interquartile Range	
		Skewness	.687
		Kurtosis	1.334
	ECC	Mean	.000000
		95% Confidence Interval for Mean	Lower Bound
			Upper Bound
		5% Trimmed Mean	
		Median	
		Variance	
		Std. Deviation	
		Minimum	
		Maximum	
		Range	
		Interquartile Range	
		Skewness	.

Descriptives

Encryption condition code		Statistic	
	RSA-SHE	Kurtosis	.
		Mean	3104.00000
		95% Confidence Interval for Mean	Lower Bound 3104.00000 Upper Bound 3104.00000
		5% Trimmed Mean	3104.00000
		Median	3104.00000
		Variance	.000
		Std. Deviation	.000000
		Minimum	3104.000
		Maximum	3104.000
		Range	.000
		Interquartile Range	.000
		Skewness	.
		Kurtosis	.
Measured encryption time in seconds	Unencrypted	Mean	.00000000
		95% Confidence Interval for Mean	Lower Bound .00000000 Upper Bound .00000000
		5% Trimmed Mean	.00000000
		Median	.00000000
		Variance	.000
		Std. Deviation	.00000000
		Minimum	.000000
		Maximum	.000000
		Range	.000000
		Interquartile Range	.000000
		Skewness	.
		Kurtosis	.
	ECC	Mean	.06399869
		95% Confidence Interval for Mean	Lower Bound .04215210 Upper Bound .08584527
		5% Trimmed Mean	.06151249
		Median	.04456080
		Variance	.002
		Std. Deviation	.039449822
		Minimum	.030559
		Maximum	.142190
		Range	.111631
		Interquartile Range	.063948
		Skewness	1.025

Descriptives

Encryption condition code		Std. Error	
	RSA-SHE	Kurtosis	.
		Mean	.000000
		95% Confidence Interval for Mean	Lower Bound
			Upper Bound
		5% Trimmed Mean	
		Median	
		Variance	
		Std. Deviation	
		Minimum	
		Maximum	
		Range	
		Interquartile Range	
		Skewness	.
		Kurtosis	.
Measured encryption time in seconds	Unencrypted	Mean	.000000000
		95% Confidence Interval for Mean	Lower Bound
			Upper Bound
		5% Trimmed Mean	
		Median	
		Variance	
		Std. Deviation	
		Minimum	
		Maximum	
		Range	
		Interquartile Range	
		Skewness	.
		Kurtosis	.
	ECC	Mean	.010185900
		95% Confidence Interval for Mean	Lower Bound
			Upper Bound
		5% Trimmed Mean	
		Median	
		Variance	
		Std. Deviation	
		Minimum	
		Maximum	
		Range	
		Interquartile Range	
		Skewness	.580

Descriptives

Encryption condition code			Statistic
RSA-SHE	Kurtosis		-.575
	Mean		2.92742547
	95% Confidence Interval for Mean	Lower Bound	2.38797066
		Upper Bound	3.46688027
	5% Trimmed Mean		2.89628430
	Median		2.96863790
	Variance		.949
	Std. Deviation		.974129258
	Minimum		1.494043
	Maximum		4.921349
	Range		3.427305
	Interquartile Range		1.458802
	Skewness		.567
	Kurtosis		-.123

Descriptives

Encryption condition code			Std. Error
RSA-SHE	Kurtosis		1.121
	Mean		.251519093
	95% Confidence Interval for Mean	Lower Bound	
		Upper Bound	
	5% Trimmed Mean		
	Median		
	Variance		
	Std. Deviation		
	Minimum		
	Maximum		
	Range		
	Interquartile Range		
	Skewness		.580
	Kurtosis		1.121

Tests of Normality

	Encryption condition code	Kolmogorov-Smirnov ^a			Shapiro-...
		Statistic	df	Sig.	Statistic
Output file size in kilobytes	Unencrypted	.133	10	.200 [*]	.932
	ECC	.350	15	<.001	.779
	RSA-SHE	.147	15	.200 [*]	.958
Average row length in bytes	Unencrypted	.133	10	.200 [*]	.932
	ECC	.	15	.	.
	RSA-SHE	.	15	.	.
Measured encryption time in seconds	Unencrypted	.	10	.	.
	ECC	.335	15	<.001	.783
	RSA-SHE	.126	15	.200 [*]	.961

Tests of Normality

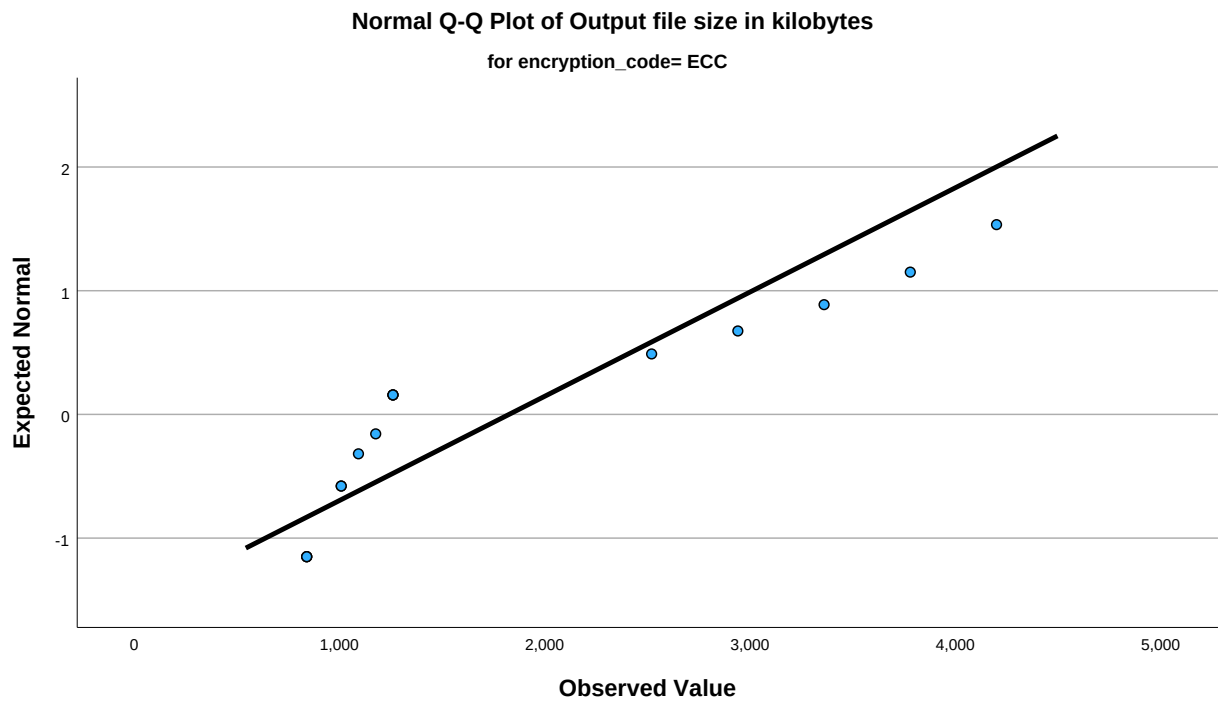
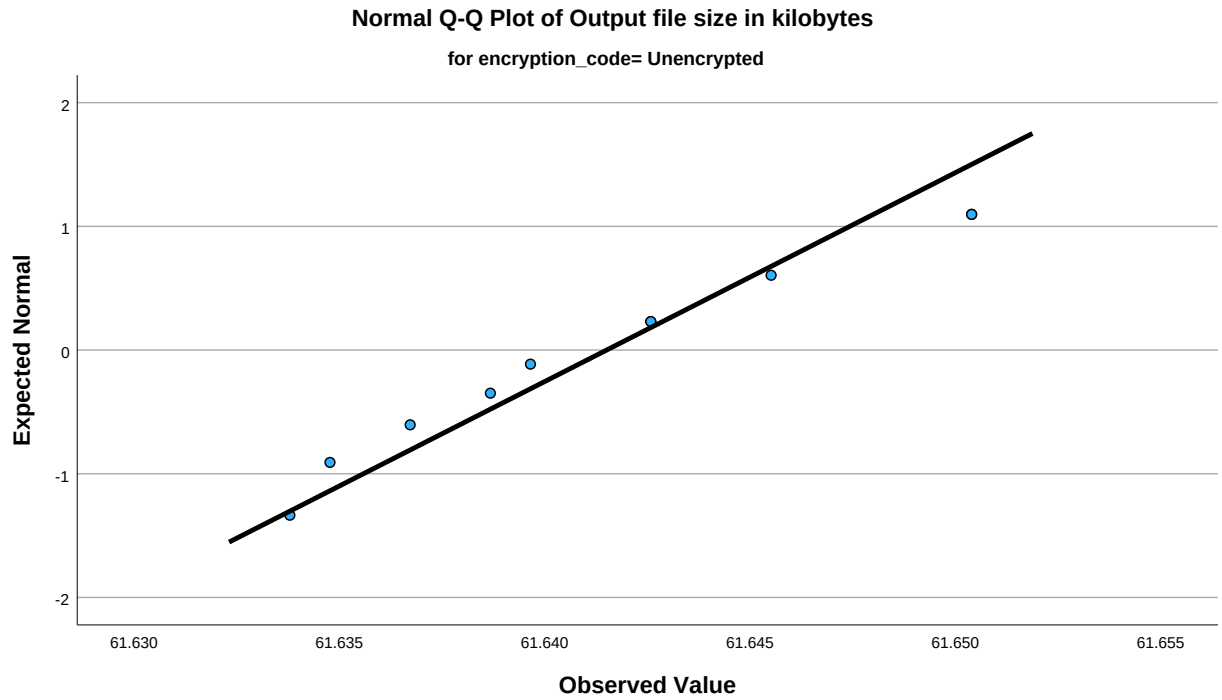
	Encryption condition code	Shapiro-Wilk	
		df	Sig.
Output file size in kilobytes	Unencrypted	10	.463
	ECC	15	.002
	RSA-SHE	15	.654
Average row length in bytes	Unencrypted	10	.464
	ECC	15	.
	RSA-SHE	15	.
Measured encryption time in seconds	Unencrypted	10	.
	ECC	15	.002
	RSA-SHE	15	.711

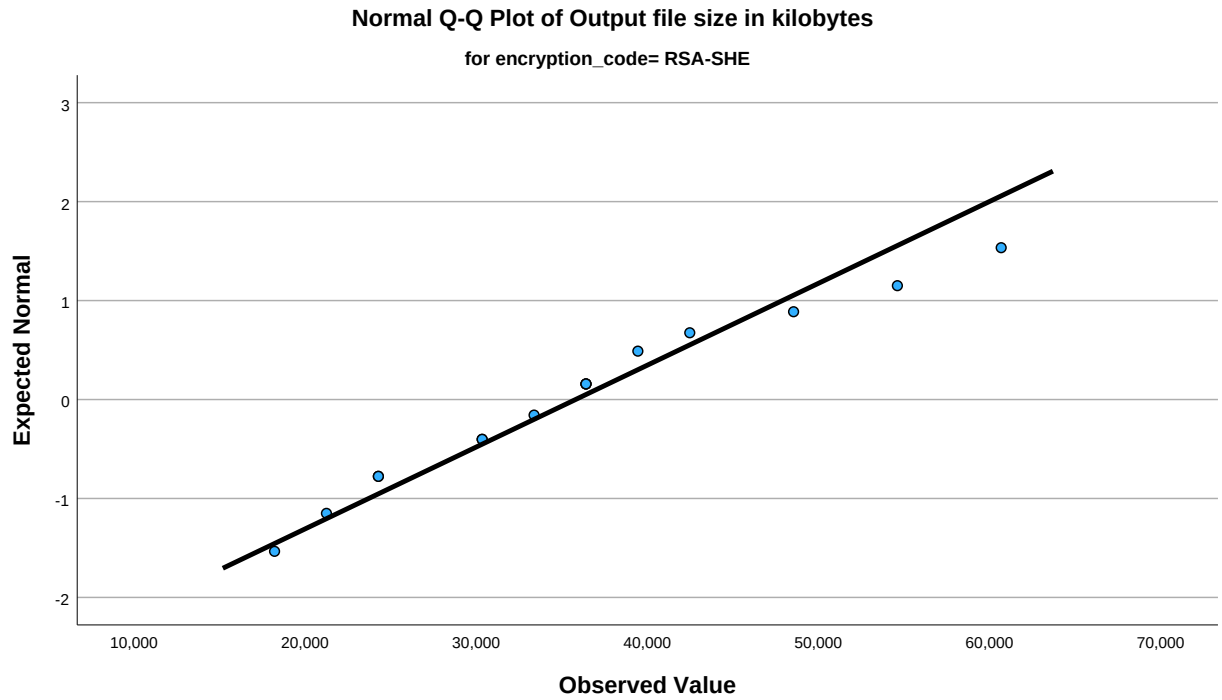
*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

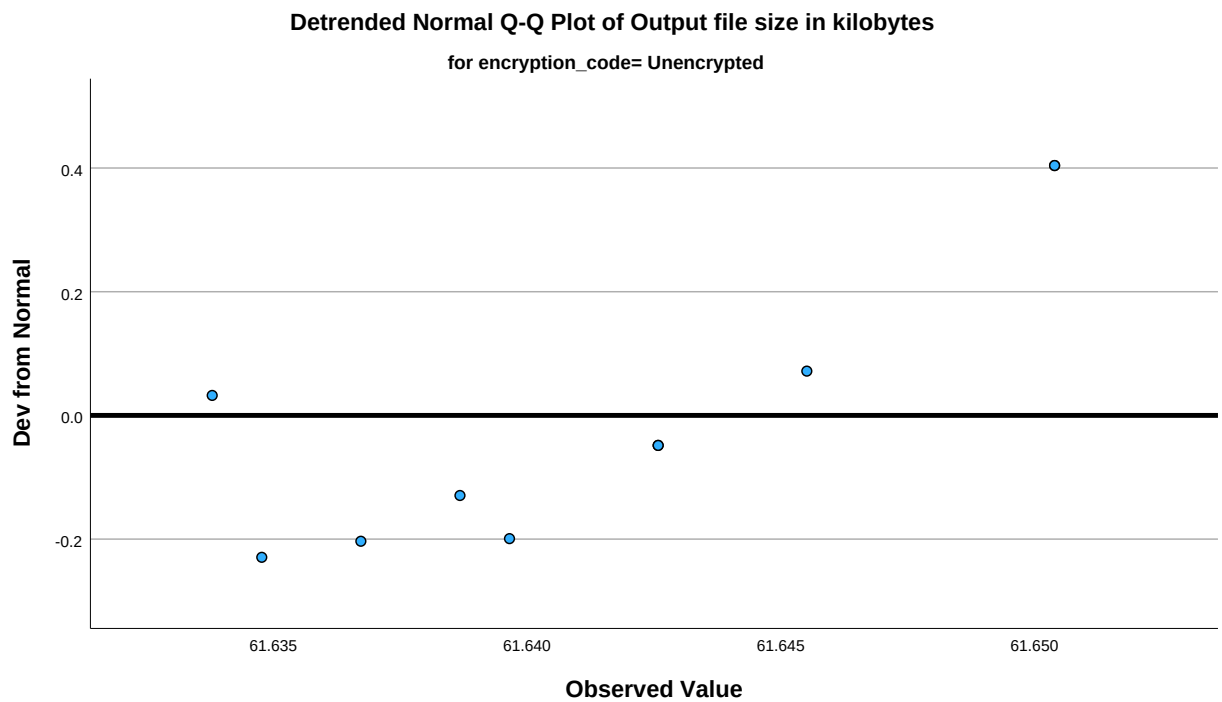
Output file size in kilobytes

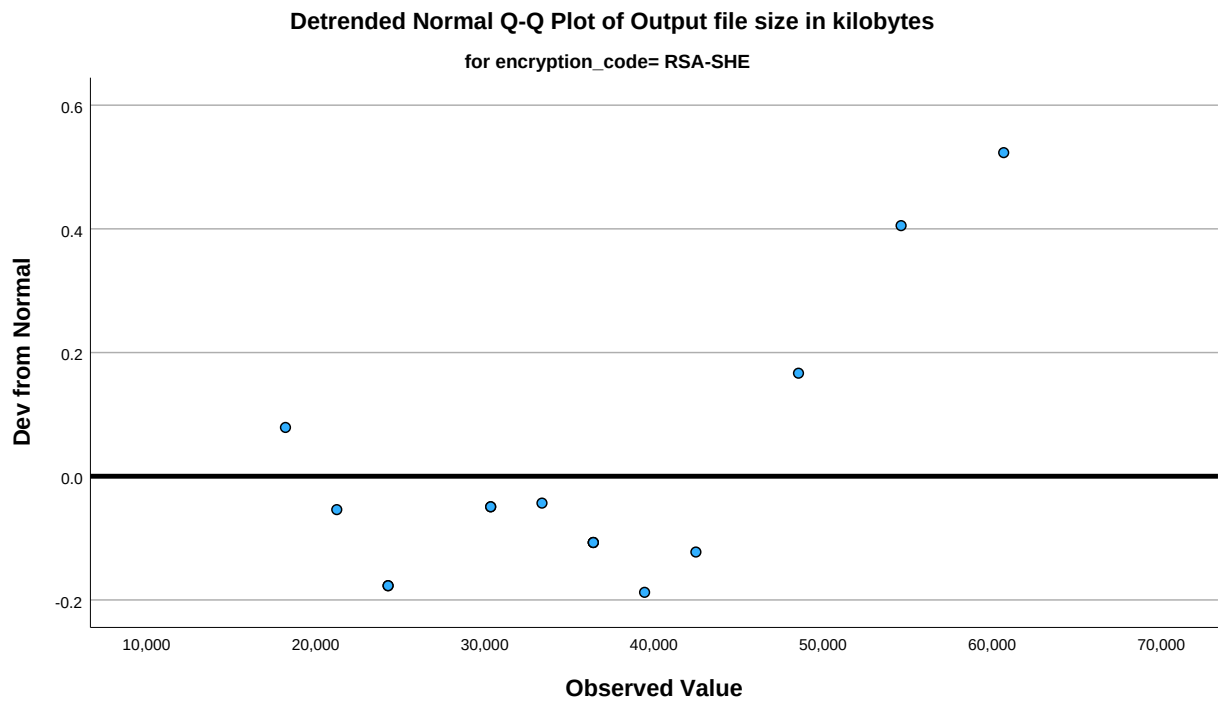
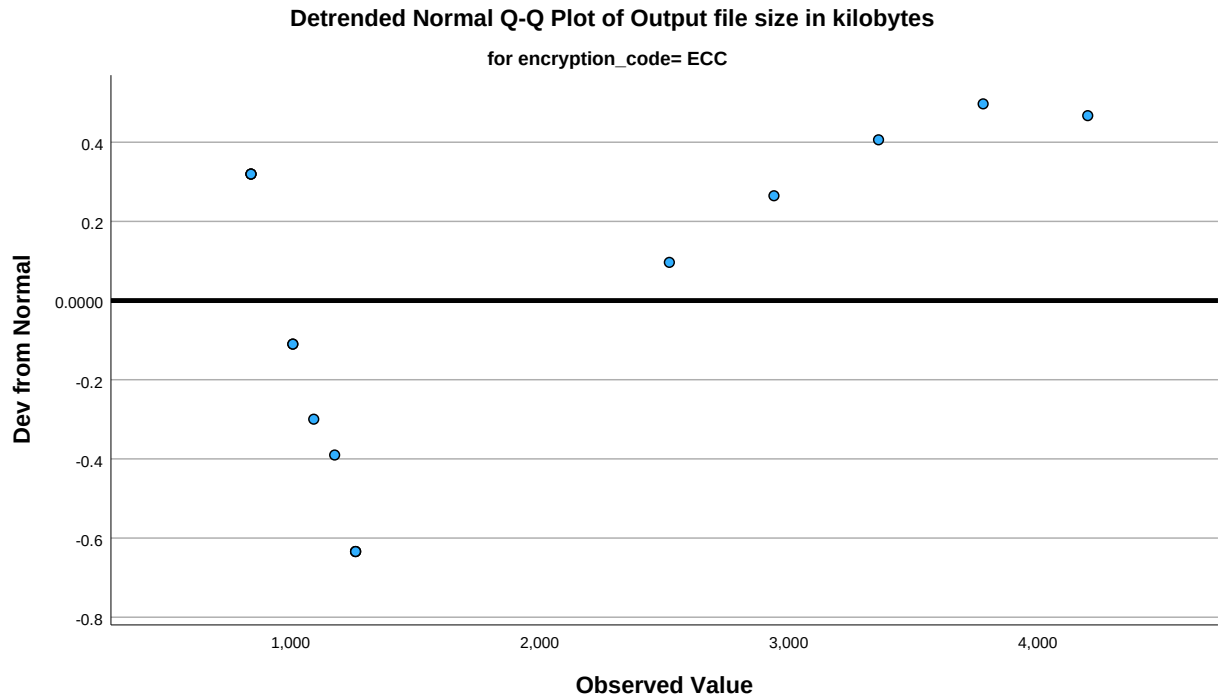
Normal Q-Q Plots

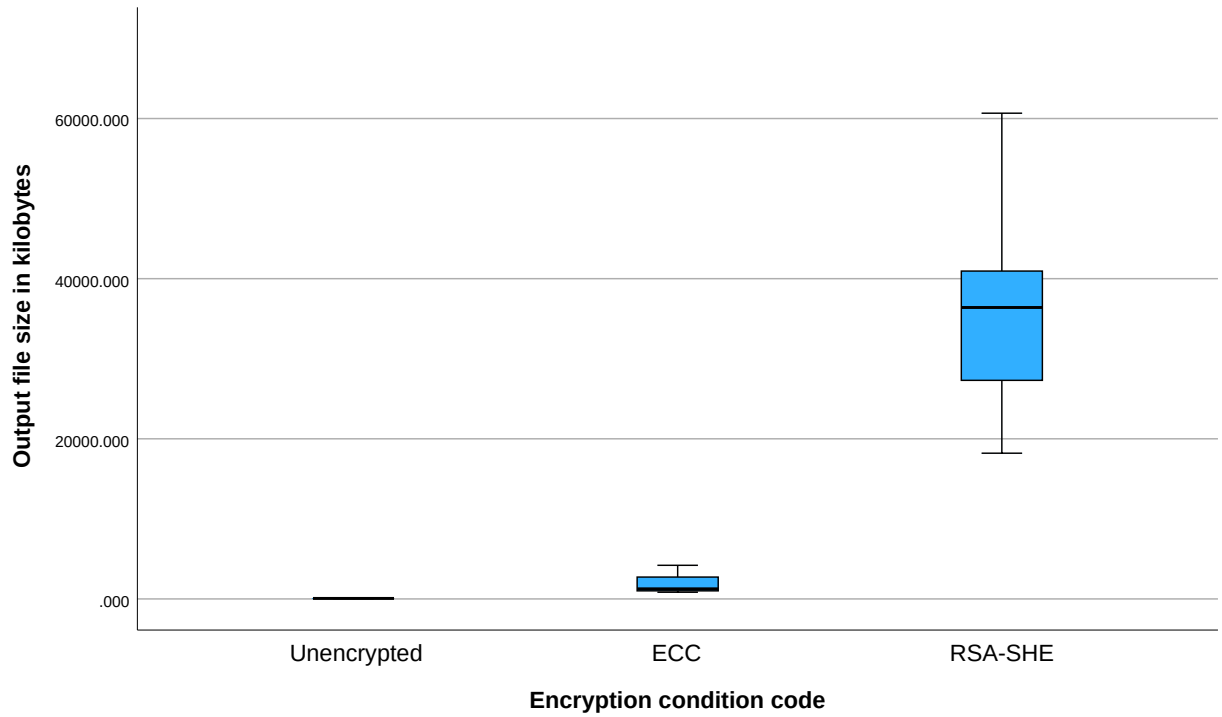




Detrended Normal Q-Q Plots

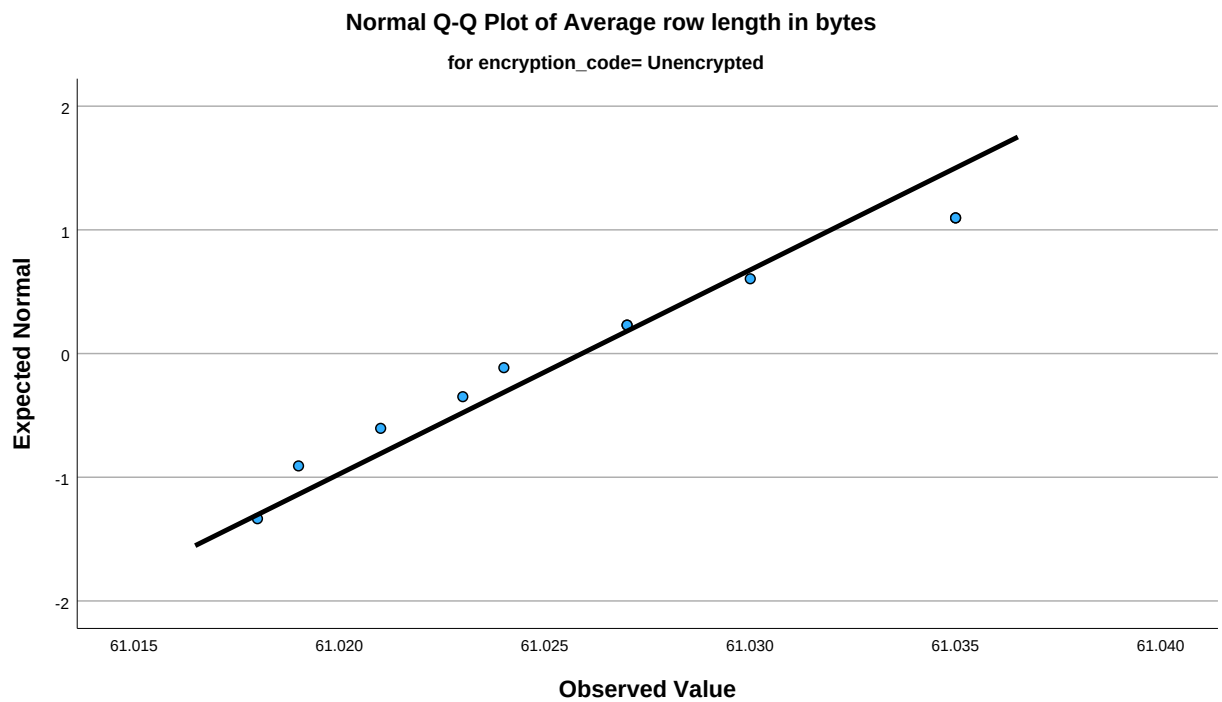


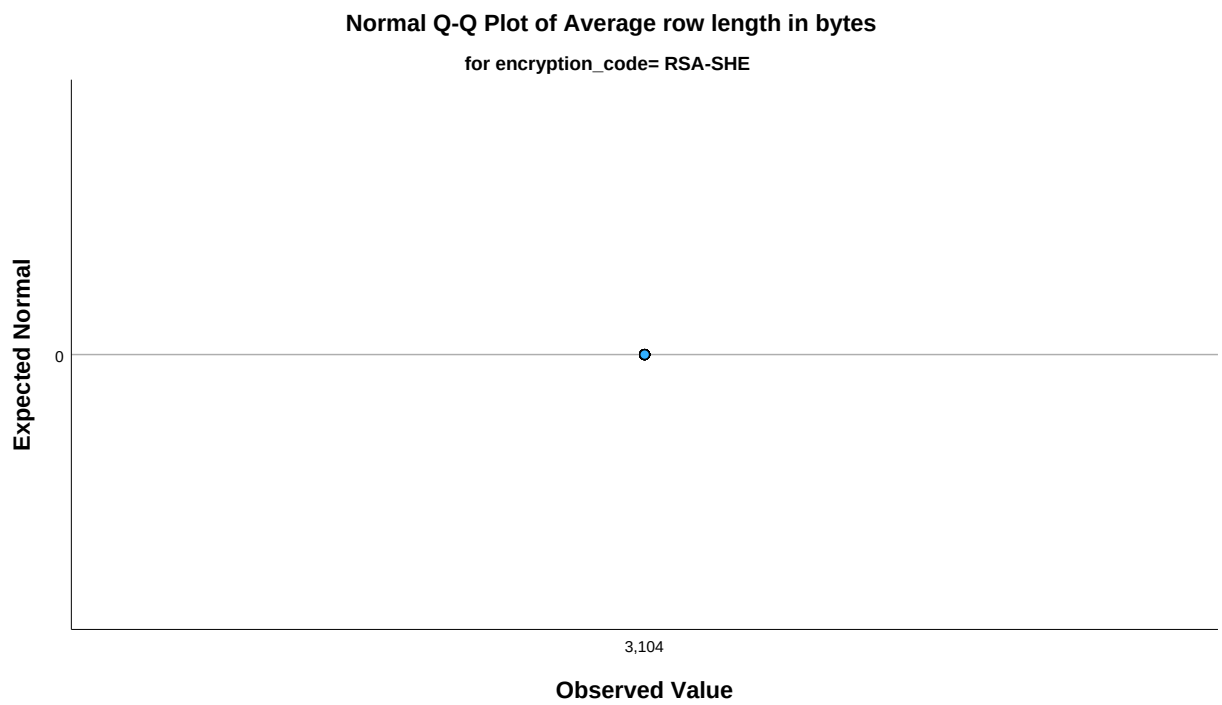
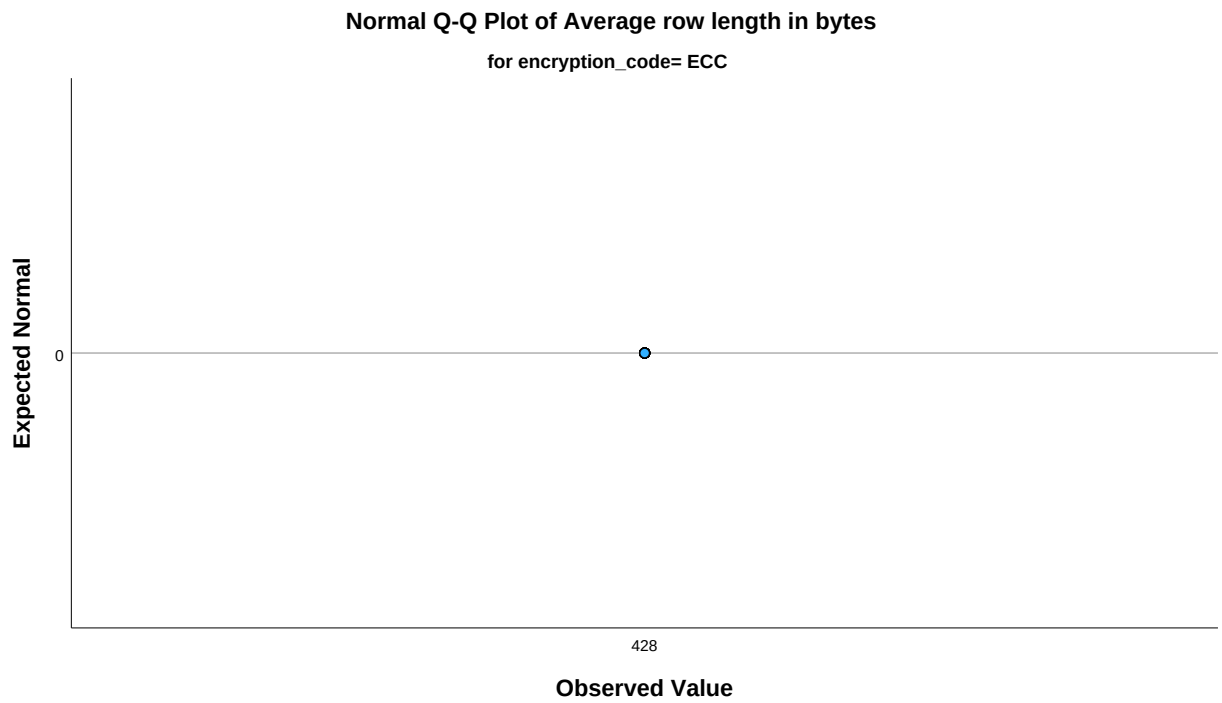




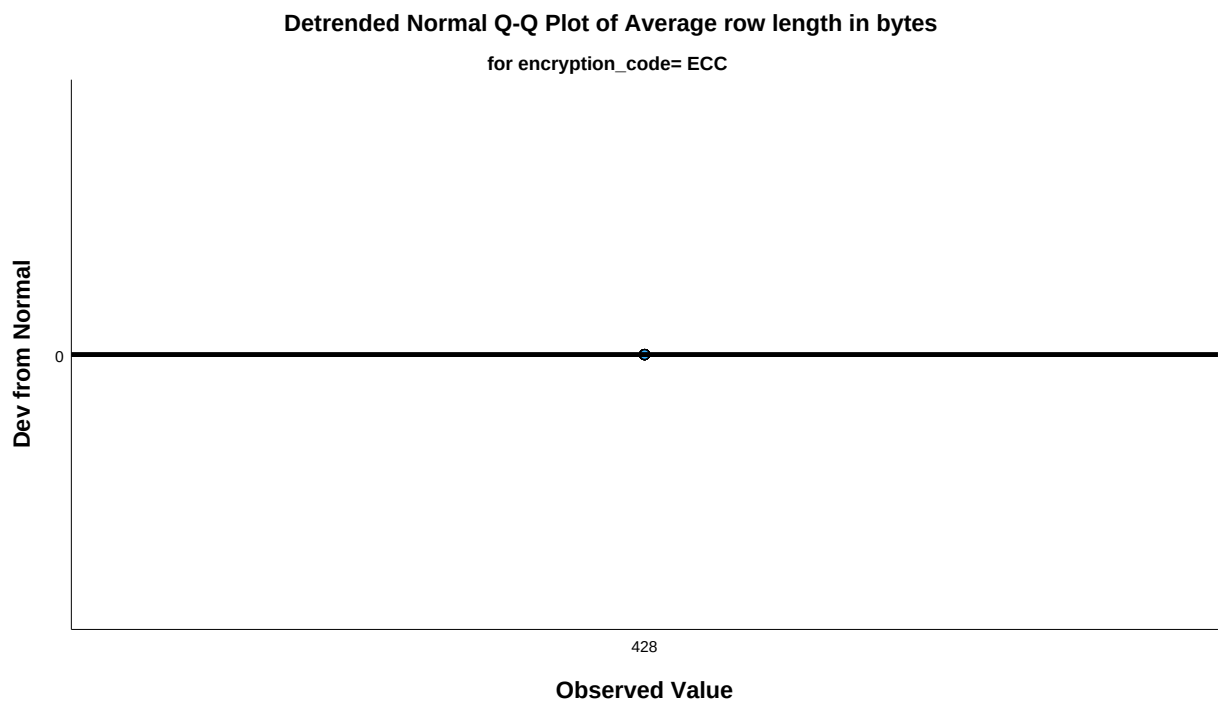
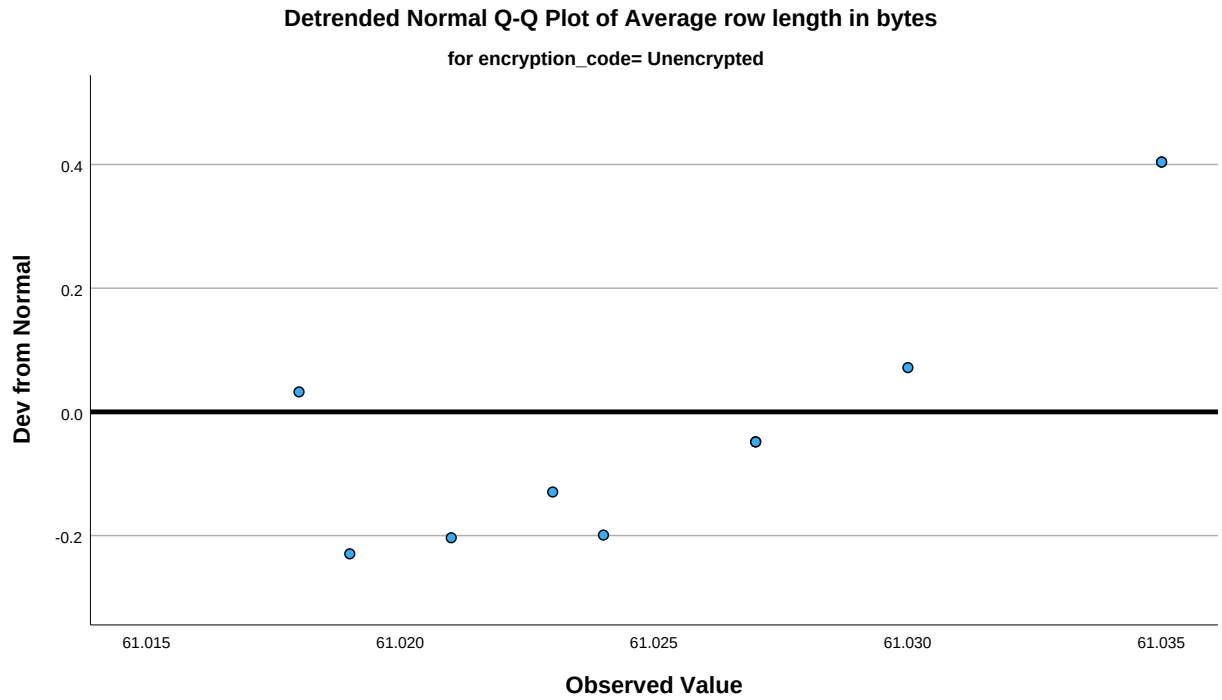
Average row length in bytes

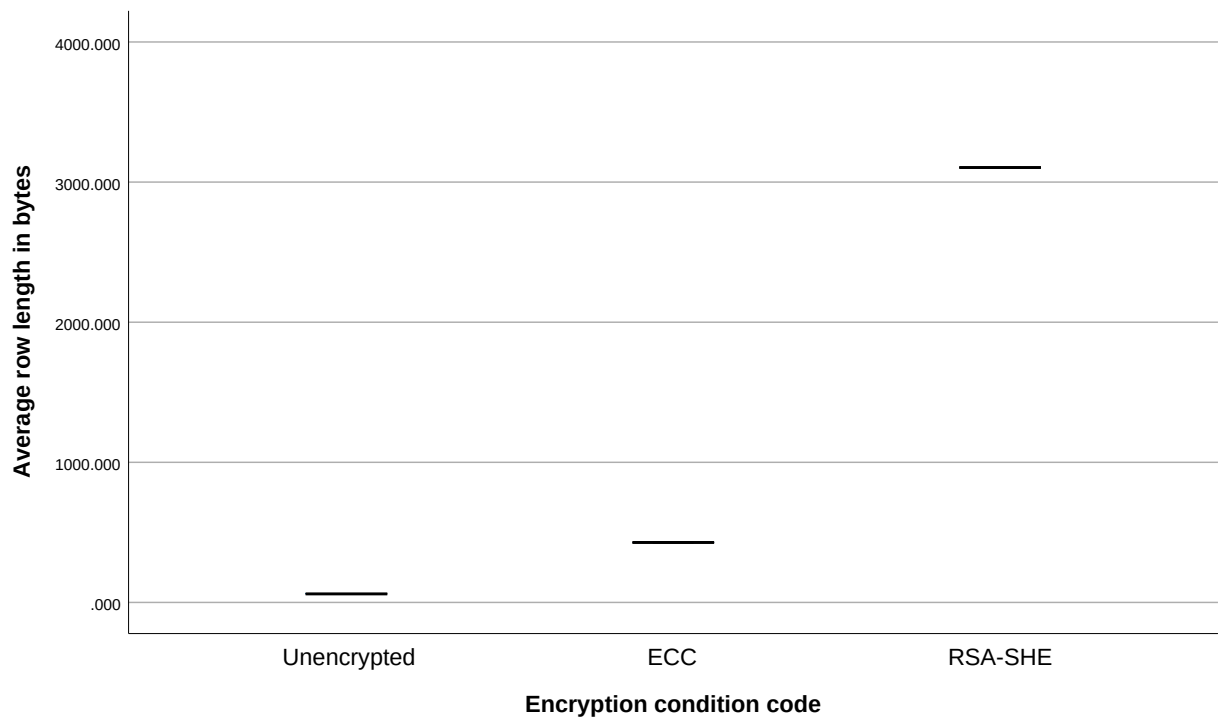
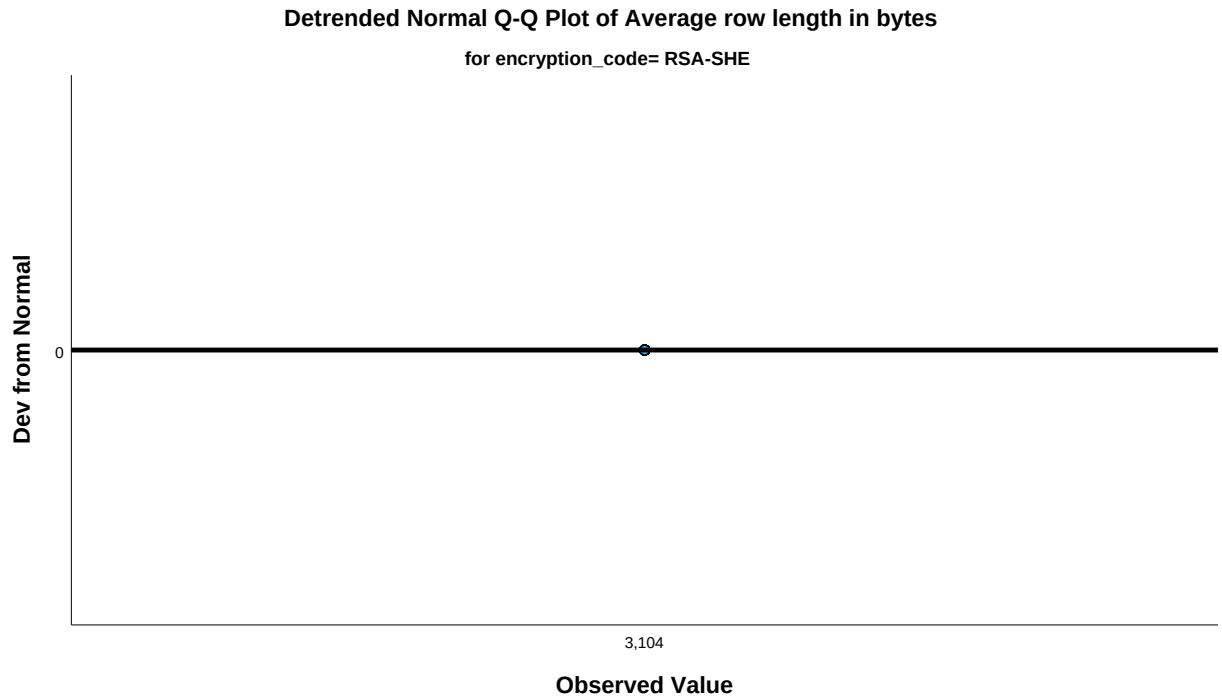
Normal Q-Q Plots





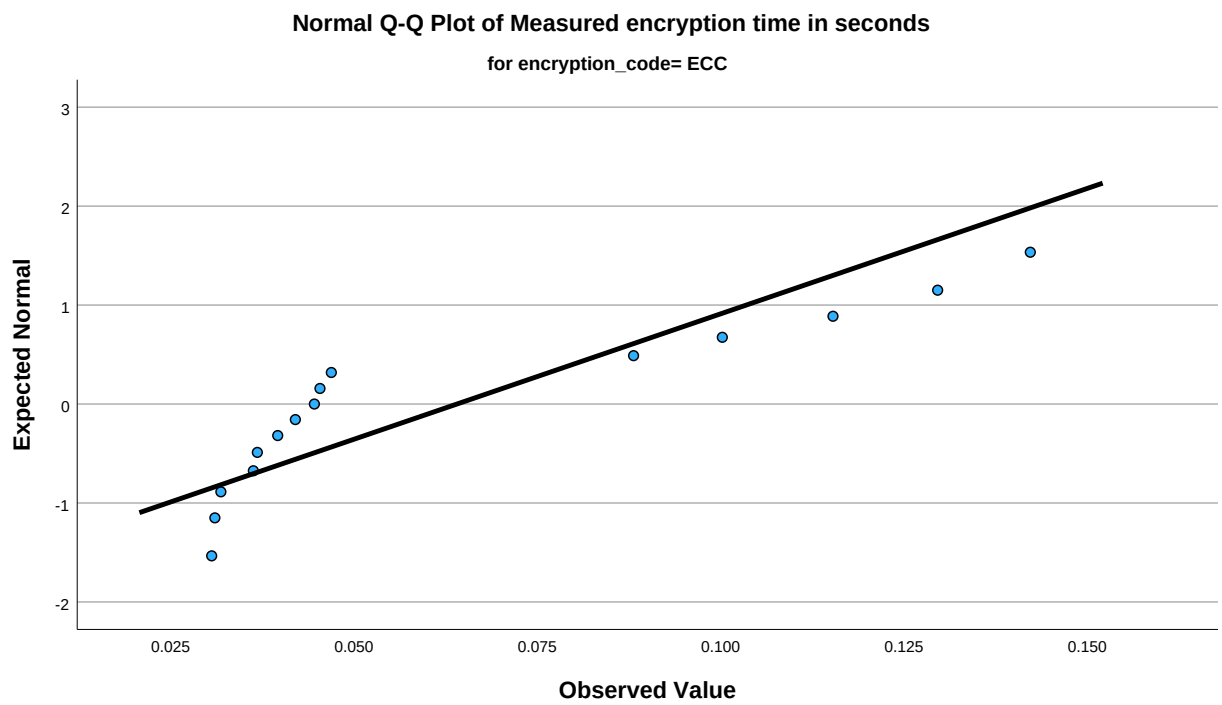
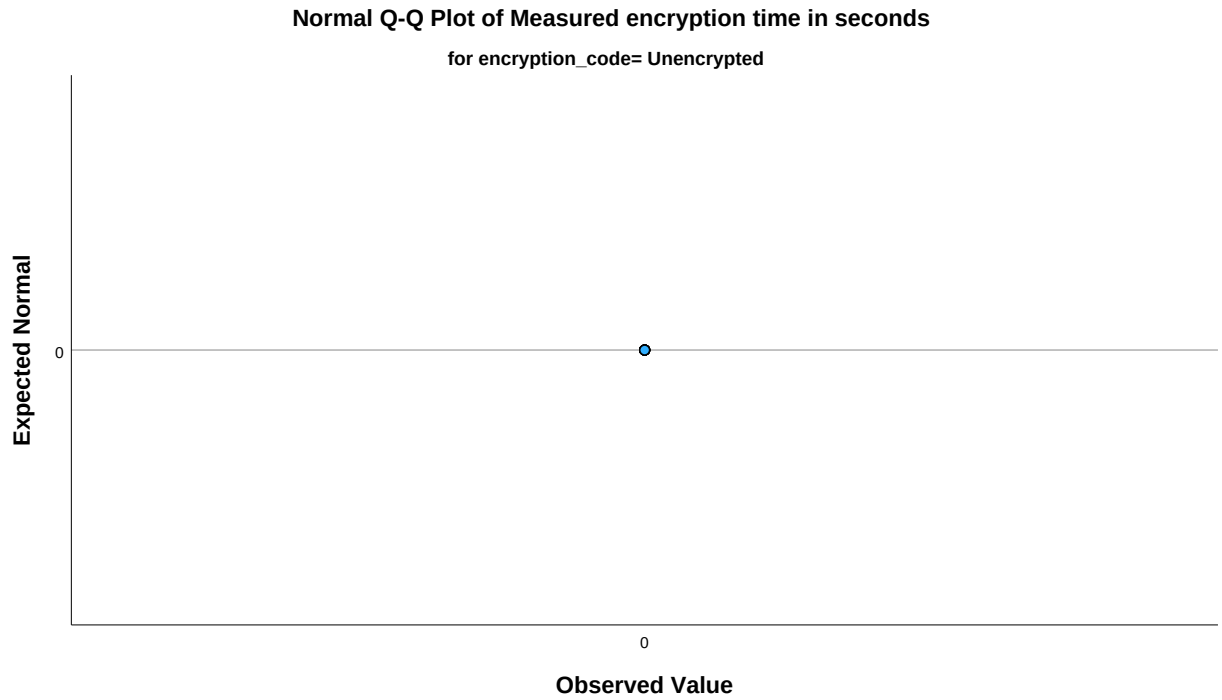
Detrended Normal Q-Q Plots

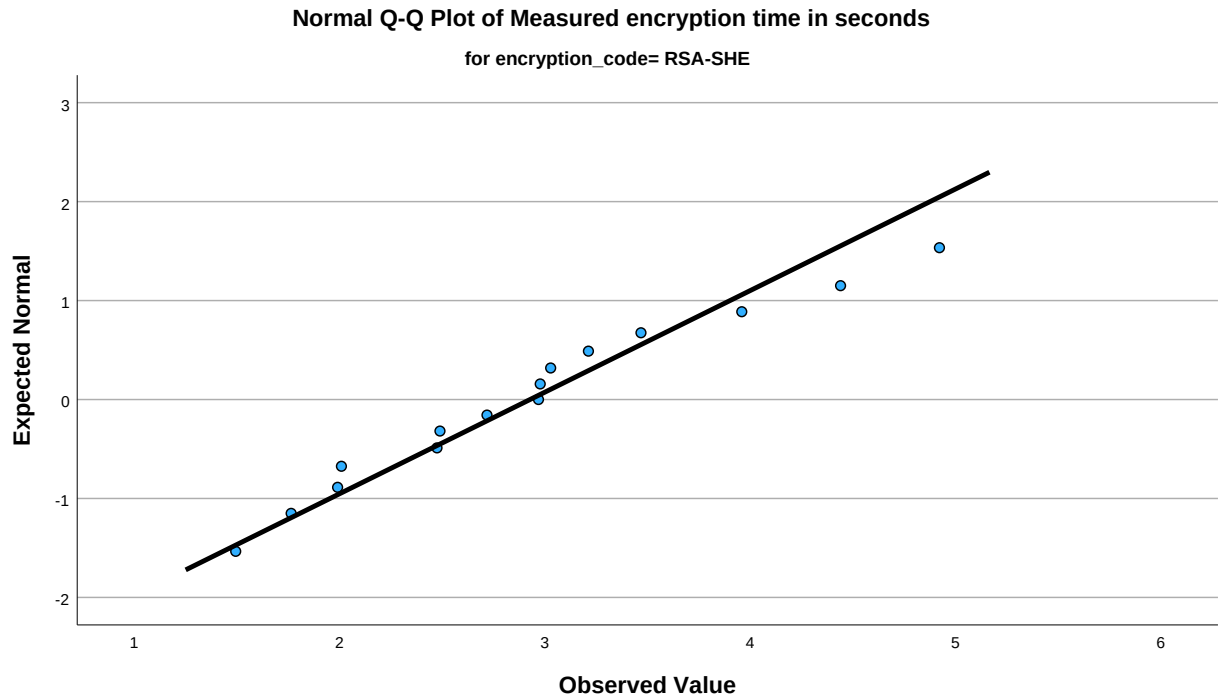




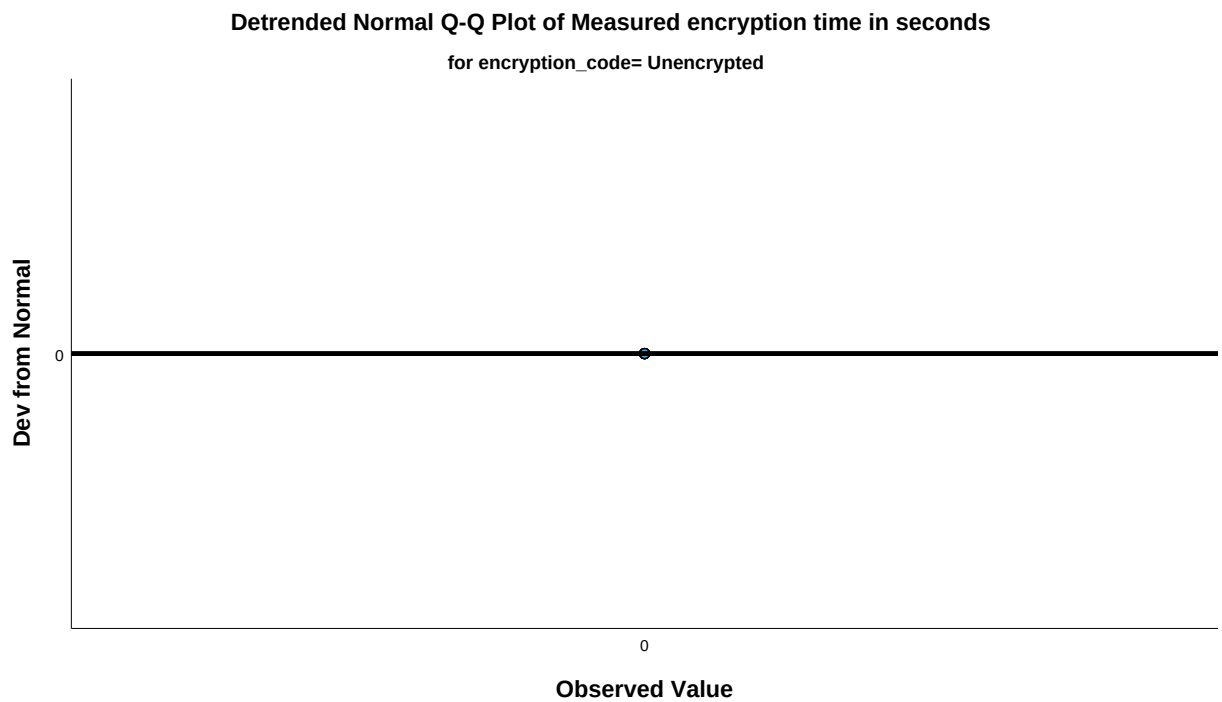
Measured encryption time in seconds

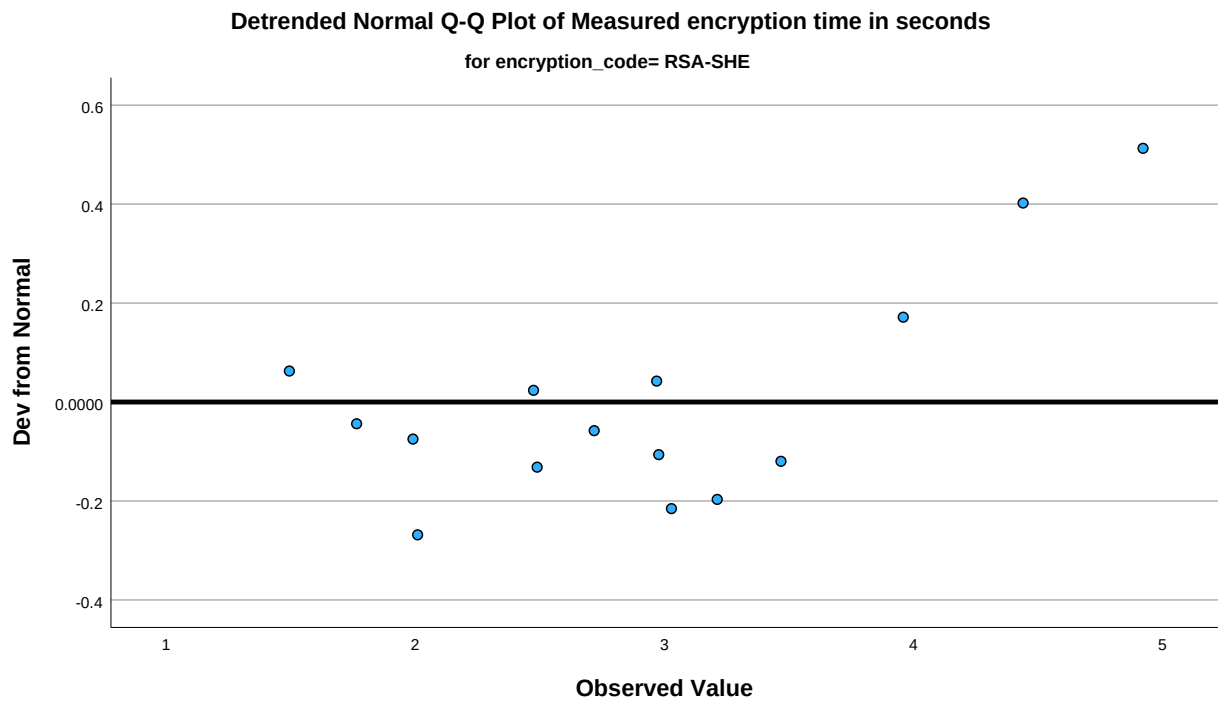
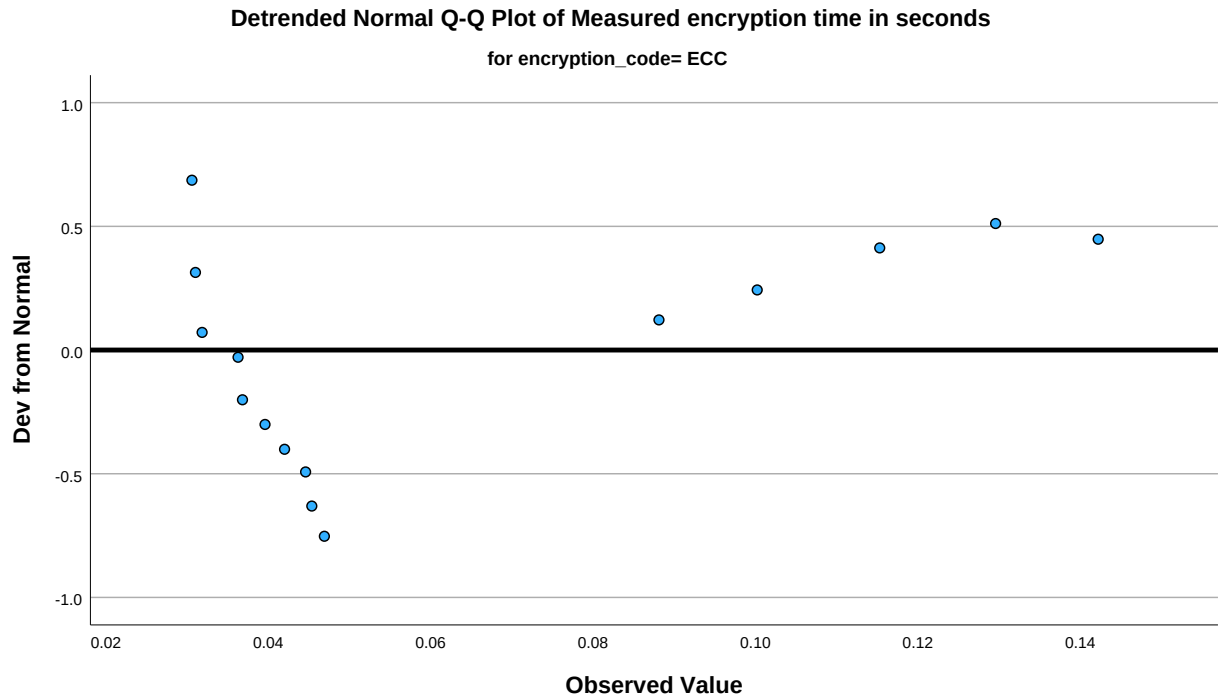
Normal Q-Q Plots

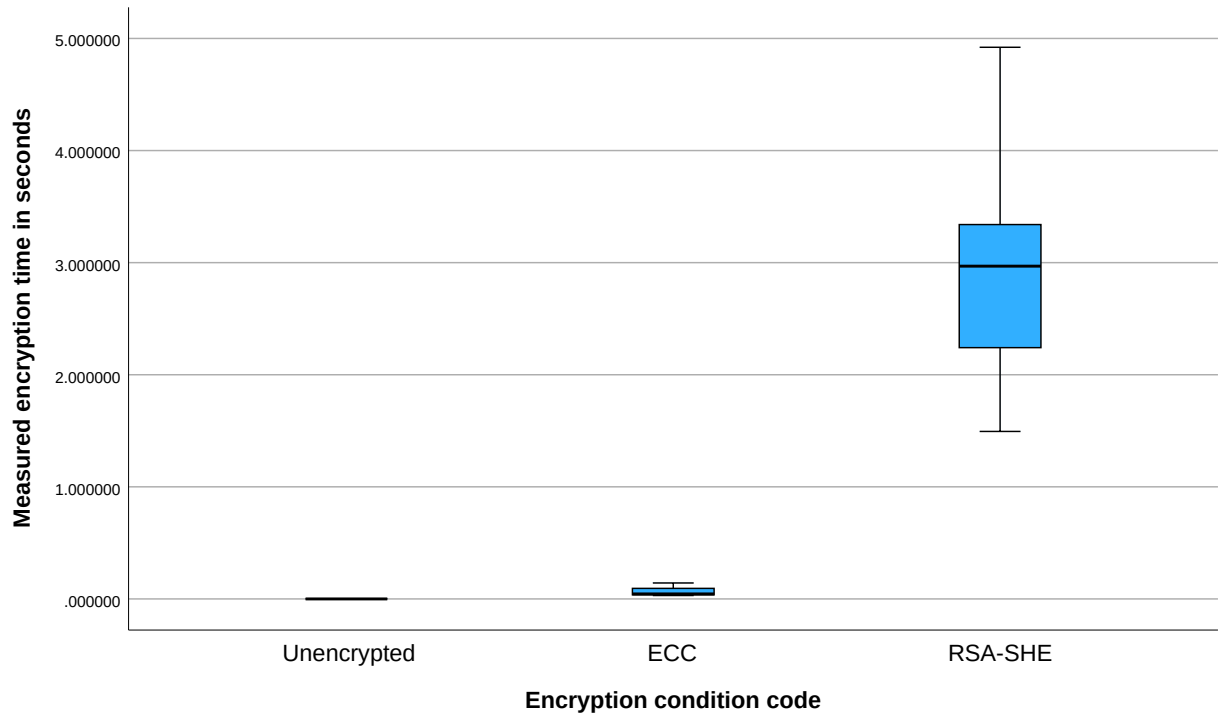




Detrended Normal Q-Q Plots







Oneway

Descriptives

Output file size in kilobytes

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean	
					Lower Bound	Upper Bound
Unencrypted	10	61.64150	.005922	.001873	61.63727	61.64574
ECC	15	1825.35319	1186.218687	306.280348	1168.44718	2482.25920
RSA-SHE	15	35791.88965	12089.403372	3121.470528	29097.00121	42486.77808
Total	40	14121.87644	18504.789715	2925.864156	8203.75758	20039.99530

Descriptives

Output file size in kilobytes

	Minimum	Maximum
Unencrypted	61.634	61.650
ECC	839.937	4199.312
RSA-SHE	18199.312	60664.155
Total	61.634	60664.155

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2
Output file size in kilobytes	Based on Mean	15.965	2	37
	Based on Median	15.664	2	37
	Based on Median and with adjusted df	15.664	2	14.486
	Based on trimmed mean	16.198	2	37

Tests of Homogeneity of Variances

		Sig.
Output file size in kilobytes	Based on Mean	<.001
	Based on Median	<.001
	Based on Median and with adjusted df	<.001
	Based on trimmed mean	<.001

ANOVA

Output file size in kilobytes

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	11288811412	2	5644405705.9	101.093	<.001
Within Groups	2065851041.2	37	55833811.923		
Total	13354662453	39			

Robust Tests of Equality of Means

Output file size in kilobytes

	Statistic ^a	df1	df2	Sig.
Welch	79.262	2	18.667	<.001
Brown-Forsythe	122.404	2	14.270	<.001

a. Asymptotically F distributed.

Post Hoc Tests

Multiple Comparisons

Dependent Variable: Output file size in kilobytes

Games-Howell

(I) Encryption condition code	(J) Encryption condition code	Mean Difference (I-J)	Std. Error	Sig.
Unencrypted	ECC	-1763.711686*	306.280348	<.001
	RSA-SHE	-35730.24814*	3121.470528	<.001
ECC	Unencrypted	1763.711686*	306.280348	<.001
	RSA-SHE	-33966.53646*	3136.460730	<.001
RSA-SHE	Unencrypted	35730.248144*	3121.470528	<.001
	ECC	33966.536458*	3136.460730	<.001

Multiple Comparisons

Dependent Variable: Output file size in kilobytes

Games-Howell

(I) Encryption condition code	(J) Encryption condition code	95% Confidence Interval	
		Lower Bound	Upper Bound
Unencrypted	ECC	-2565.33327	-962.09010
	RSA-SHE	-43900.01213	-27560.48416
ECC	Unencrypted	962.09010	2565.33327
	RSA-SHE	-42157.85243	-25775.22048
RSA-SHE	Unencrypted	27560.48416	43900.01213
	ECC	25775.22048	42157.85243

*. The mean difference is significant at the .05 level.

Oneway

Descriptives

Average row length in bytes

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean	
					Lower Bound	Upper Bound
Unencrypted	10	61.02590	.006064	.001917	61.02156	61.03024
ECC	15	428.00000	.000000	.000000	428.00000	428.00000
RSA-SHE	15	3104.00000	.000000	.000000	3104.00000	3104.00000
Total	40	1339.75648	1391.451420	220.007787	894.74872	1784.76423

Descriptives

Average row length in bytes

	Minimum	Maximum
Unencrypted	61.018	61.035
ECC	428.000	428.000
RSA-SHE	3104.000	3104.000
Total	61.018	3104.000

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2
Average row length in bytes	Based on Mean	36.689	2	37
	Based on Median	36.054	2	37
	Based on Median and with adjusted df	36.054	2	9.000
	Based on trimmed mean	36.671	2	37

Tests of Homogeneity of Variances

		Sig.
Average row length in bytes	Based on Mean	<.001
	Based on Median	<.001
	Based on Median and with adjusted df	<.001
	Based on trimmed mean	<.001

ANOVA

Average row length in bytes

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	75509345.113	2	37754672.556	4.222E+12	<.001
Within Groups	.000	37	.000		
Total	75509345.113	39			

Robust Tests of Equality of Means^b

Average row length in bytes

	Statistic ^a	df1	df2	Sig.
Welch	.	.	.	.
Brown-Forsythe	.	.	.	.

a. Asymptotically F distributed.

b. Robust tests of equality of means cannot be performed for Average row length in bytes because at least one group has 0 variance.

Oneway

Descriptives

Measured encryption time in seconds

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean	
					Lower Bound	Upper Bound
Unencrypted	10	.00000000	.000000000	.000000000	.00000000	.00000000
ECC	15	.06399869	.039449822	.010185900	.04215210	.08584527
RSA-SHE	15	2.92742547	.974129258	.251519093	2.38797066	3.46688027
Total	40	1.12178406	1.532381538	.242290795	.63170467	1.61186345

Descriptives

Measured encryption time in seconds

	Minimum	Maximum
Unencrypted	.000000	.000000
ECC	.030559	.142190
RSA-SHE	1.494043	4.921349
Total	.000000	4.921349

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2
Measured encryption time in seconds	Based on Mean	17.781	2	37
	Based on Median	17.480	2	37
	Based on Median and with adjusted df	17.480	2	14.086
	Based on trimmed mean	18.017	2	37

Tests of Homogeneity of Variances

		Sig.
Measured encryption time in seconds	Based on Mean	<.001
	Based on Median	<.001
	Based on Median and with adjusted df	<.001
	Based on trimmed mean	<.001

ANOVA

Measured encryption time in seconds

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	78.273	2	39.136	108.820	<.001
Within Groups	13.307	37	.360		
Total	91.580	39			

Robust Tests of Equality of Means^b

Measured encryption time in seconds

	Statistic ^a	df1	df2	Sig.
Welch	.	.	.	.
Brown-Forsythe	.	.	.	.

a. Asymptotically F distributed.

b. Robust tests of equality of means cannot be performed for Measured encryption time in seconds because at least one group has 0 variance.

Nonparametric Tests

Hypothesis Test Summary

	Null Hypothesis	Test	Sig. ^{a,b}
1	The distribution of Average row length in bytes is the same across categories of Encryption condition code.	Independent-Samples Kruskal-Wallis Test	<.001

Hypothesis Test Summary

	Decision
1	Reject the null hypothesis.

- a. The significance level is .050.
- b. Asymptotic significance is displayed.

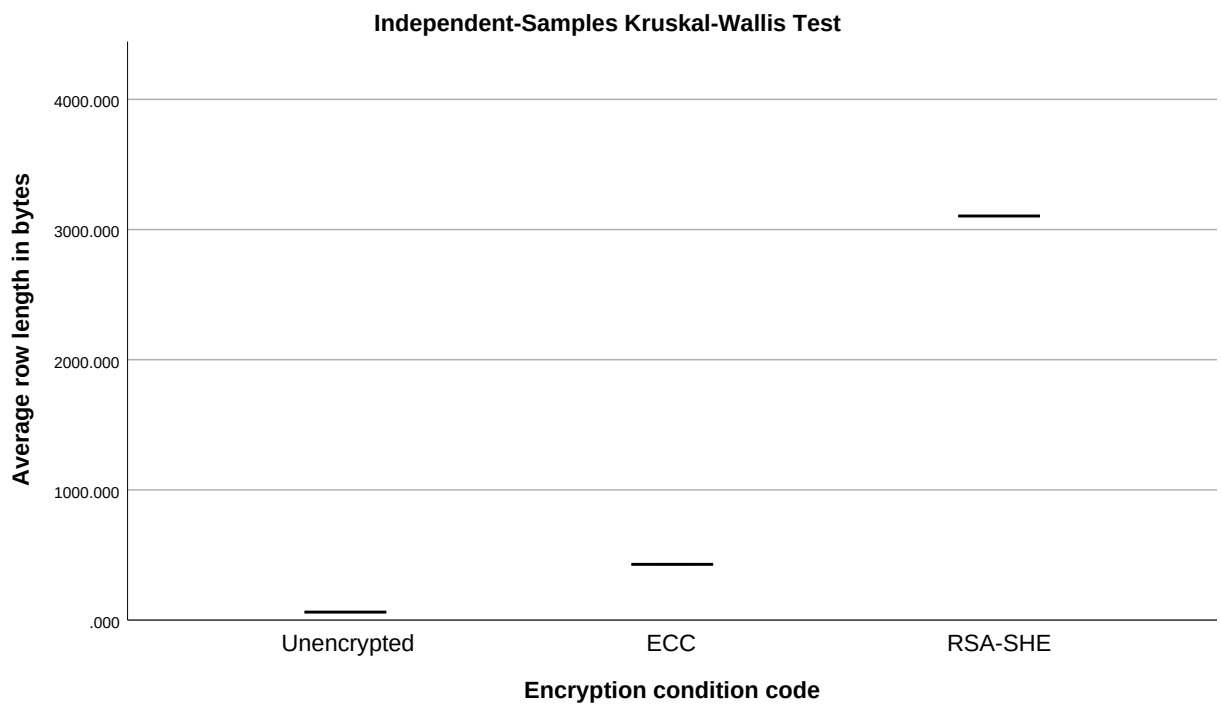
Independent-Samples Kruskal-Wallis Test

Average row length in bytes across Encryption condition code

Independent-Samples Kruskal-Wallis Test Summary

Total N	40
Test Statistic	38.334 ^a
Degree Of Freedom	2
Asymptotic Sig.(2-sided test)	<.001

- a. The test statistic is adjusted for ties.



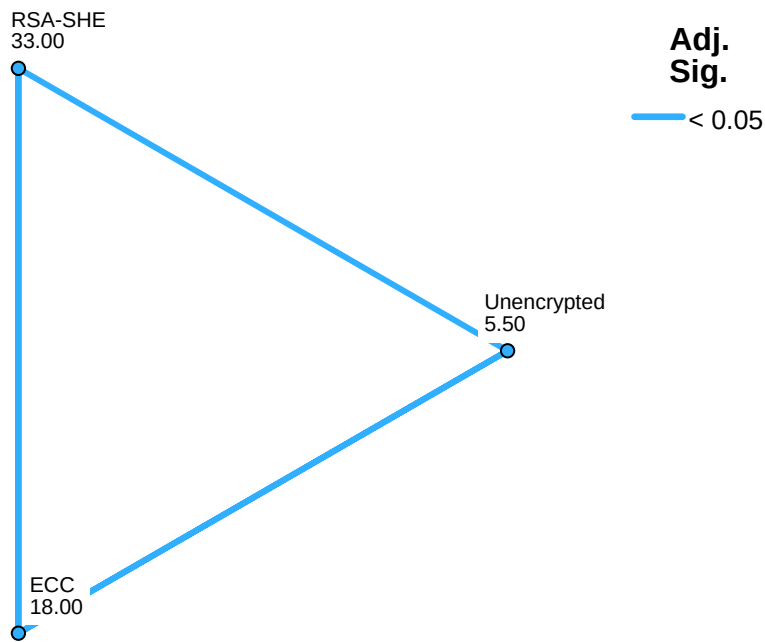
Pairwise Comparisons of Encryption condition code

Sample 1-Sample 2	Test Statistic	Std. Error	Std. Test Statistic	Sig.	Adj. Sig. ^a
Unencrypted-ECC	-12.500	4.514	-2.769	.006	.017
Unencrypted-RSA-SHE	-27.500	4.514	-6.092	<.001	.000
ECC-RSA-SHE	-15.000	4.038	-3.715	<.001	.001

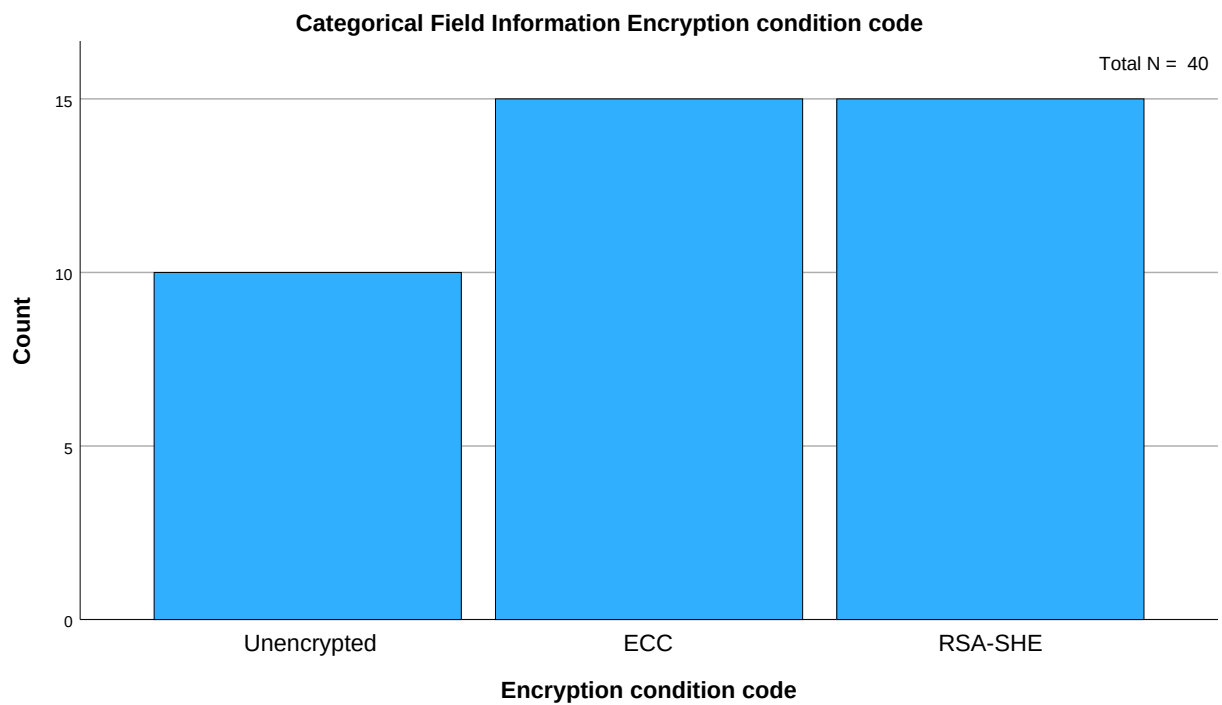
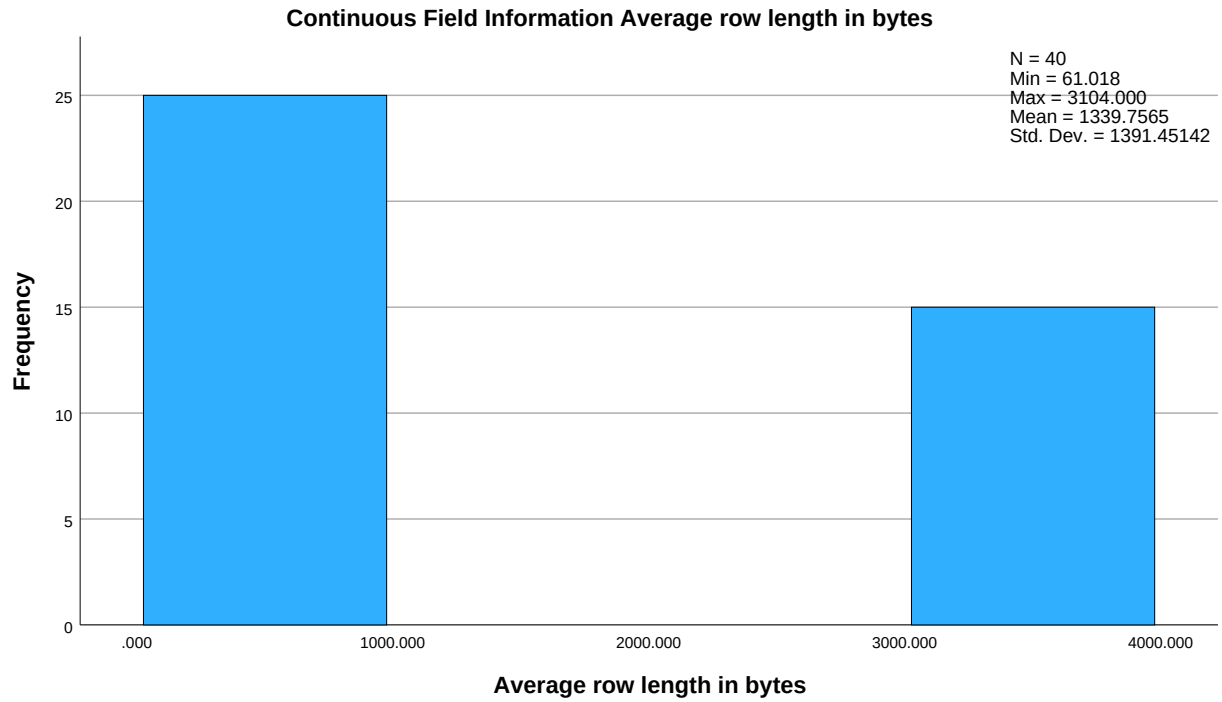
Each row tests the null hypothesis that the Sample 1 and Sample 2 distributions are the same. Asymptotic significances (2-sided tests) are displayed. The significance level is .050.

a. Significance values have been adjusted by the Bonferroni correction for multiple tests.

Pairwise Comparisons of Encryption condition code



Each node shows the sample average rank of Encryption condition code.



Nonparametric Tests

Hypothesis Test Summary

	Null Hypothesis	Test	Sig. ^{a,b}
1	The distribution of Measured encryption time in seconds is the same across categories of Encryption condition code.	Independent-Samples Kruskal-Wallis Test	<.001

Hypothesis Test Summary

	Decision
1	Reject the null hypothesis.

a. The significance level is .050.

b. Asymptotic significance is displayed.

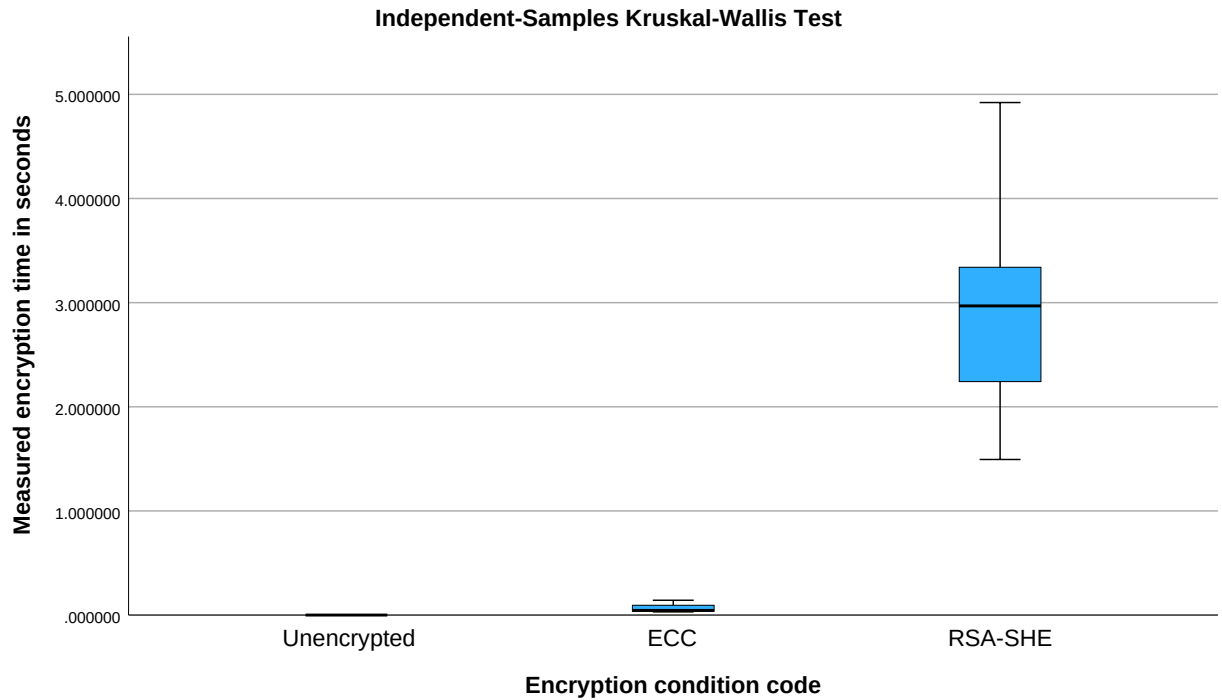
Independent-Samples Kruskal-Wallis Test

Measured encryption time in seconds across Encryption condition code

Independent-Samples Kruskal-Wallis Test Summary

Total N	40
Test Statistic	34.838 ^a
Degree Of Freedom	2
Asymptotic Sig.(2-sided test)	<.001

a. The test statistic is adjusted for ties.



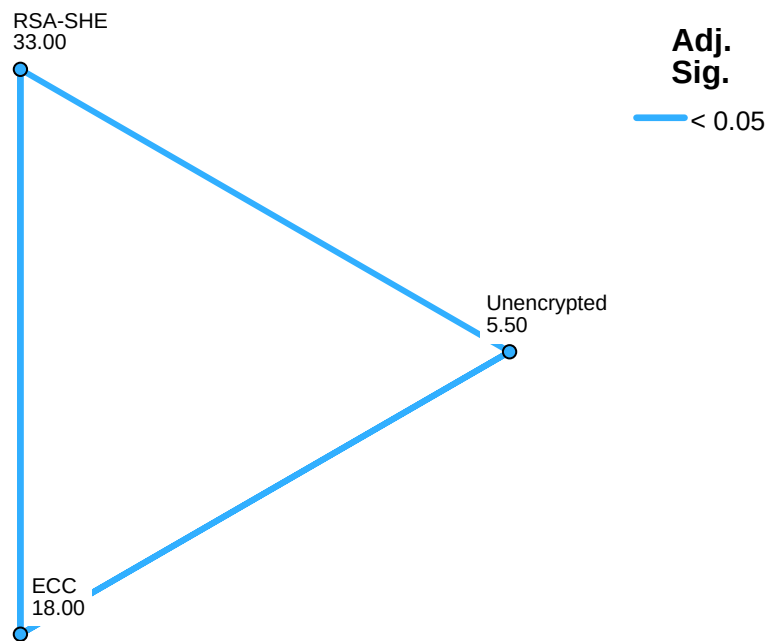
Pairwise Comparisons of Encryption condition code

Sample 1-Sample 2	Test Statistic	Std. Error	Std. Test Statistic	Sig.	Adj. Sig. ^a
Unencrypted-ECC	-12.500	4.736	-2.640	.008	.025
Unencrypted-RSA-SHE	-27.500	4.736	-5.807	<.001	.000
ECC-RSA-SHE	-15.000	4.236	-3.541	<.001	.001

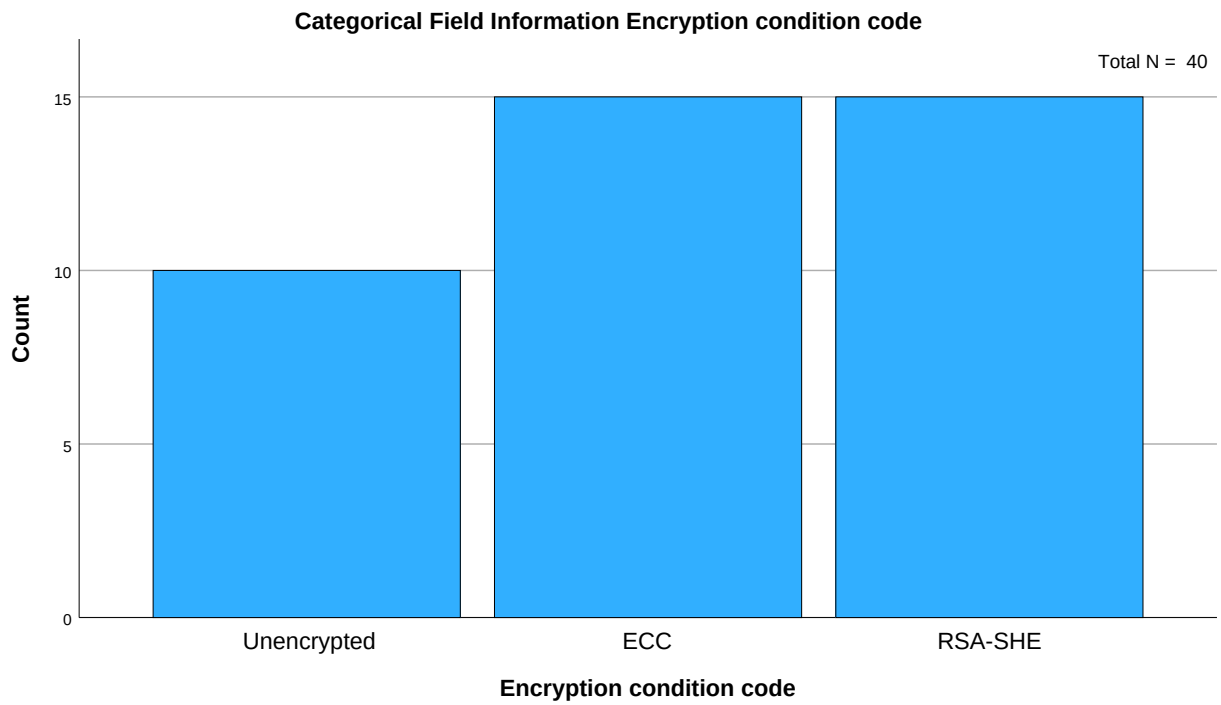
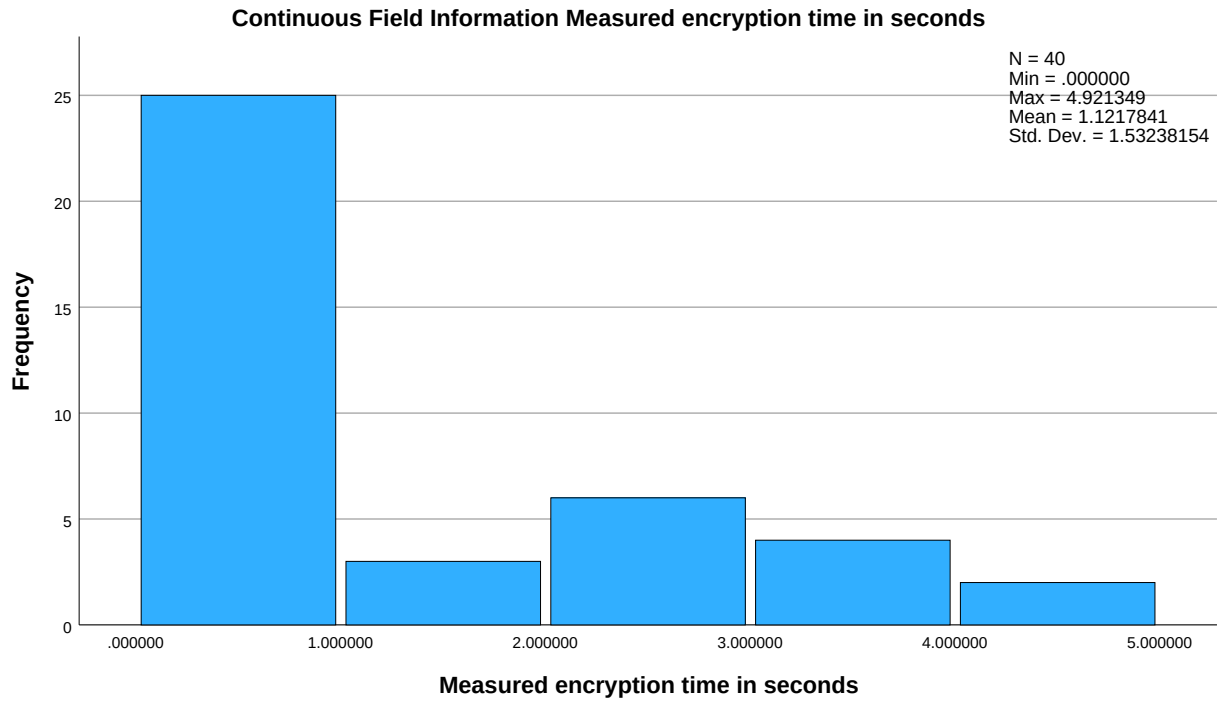
Each row tests the null hypothesis that the Sample 1 and Sample 2 distributions are the same. Asymptotic significances (2-sided tests) are displayed. The significance level is .050.

a. Significance values have been adjusted by the Bonferroni correction for multiple tests.

Pairwise Comparisons of Encryption condition code



Each node shows the sample average rank of Encryption condition code.



Means

Case Processing Summary

	Included		Cases Excluded		Total	
	N	Percent	N	Percent	N	Percent
Clear-text exposure percentage * Encryption condition code	40	100.0%	0	0.0%	40	100.0%

Report

Clear-text exposure percentage

Encryption condition code	N	Mean	Std. Deviation	Minimum	Maximum
Unencrypted	10	100.0000	.00000	100.00	100.00
ECC	15	.0000	.00000	.00	.00
RSA-SHE	15	.0000	.00000	.00	.00
Total	40	25.0000	43.85290	.00	100.00